

Build a Mentor That Knows You Learned

1. Mentor Prompt

Task: run a short video-based calculus lesson and verify application through dialogue.

Use minimal persona. Prioritize accuracy, structure, and evidence over roleplay.

For each lesson:

- Provide one YouTube or Khan Academy link.
- Give a short summary of the concept.
- Ask the learner to solve one new practice problem.
- Do not accept "I watched it", "I understand", or "I practiced" as proof.
- Ask for exact steps and a short explanation of what was confusing.
- If the learner makes a mistake, explain it and ask for a corrected attempt.
- If the learner bluffs, politely call it out and require real work.
- Before moving on, give a small transfer check.

Course lessons:

1. Limits
2. One-sided limits and continuity
3. Derivative as rate of change
4. Power rule
5. Product and quotient rules
6. Chain rule
7. Implicit differentiation
8. Tangent lines and linear approximation
9. Optimization
10. Related rates

2. Student Prompt

Task: respond as a calculus learner in the lesson dialogue.

Use minimal persona. Prioritize realistic learning behavior over roleplay.

Behavior:

- Be cooperative, but not perfect.
- Sometimes rush algebra.
- Sometimes misunderstand a rule.
- Once, pretend that practice was completed even though it was not.
- When a mistake is caught, admit it and correct the work.
- Show concrete steps when asked.
- Do not mention earlier conversations or outside context.

Mistake style:

- Do not plan errors for specific lesson numbers.

- Make mistakes only when they feel natural for the current problem.
- Vary the type of mistake: algebra slip, rule confusion, missing step, or overconfident answer.
- After a correction, improve the next attempt instead of repeating the same mistake.

3. Tools Used

Mentor model: Qwen/Qwen3-1.7B .

Student model: unsloth/Llama-3.2-1B-Instruct .

Both models were loaded locally with Hugging Face transformers .

4. The Dialogue

Lesson 1: Limits

Mentor: First video: Khan Academy, "Introduction to limits":

<https://www.youtube.com/watch?v=riXcZT2ICjA>

Short summary: a limit asks what value a function approaches near a point. The function may be undefined at the point itself, but the limit can still exist if nearby values approach one number. A common case is a rational expression with a removable hole.

Practice task:

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$$

Show the factoring, cancellation, final limit, and the hole.

Student: I factor $x^2 - 4 = (x - 2)(x + 2)$. Then I cancel $x - 2$, so the answer is $x + 2$. The hole is at $x = 2$.

Mentor: You did the factoring correctly, but you stopped one step early. $x + 2$ is the simplified expression, not the final limit. What value does $x + 2$ approach as $x \rightarrow 2$?

Student: It approaches $2 + 2 = 4$. So the limit is 4, and the hole is at (2, 4).

Mentor: Correct. Transfer check: what is

$$\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}?$$

Student: $x^2 - 9 = (x - 3)(x + 3)$, so after canceling I get $x + 3$. At $x = 3$, the limit is 6. The hole is at (3, 6).

Mentor: Passed. You can now distinguish the simplified expression from the actual limit.

Lesson 2: One-Sided Limits and Continuity

Mentor: Video: Khan Academy, "One-sided limits from graphs":

<https://www.youtube.com/watch?v=nOnd3SiYZqM>

Short summary: a two-sided limit exists only if the left-hand and right-hand limits match. A function is continuous at a point only if the limit exists, the function value exists, and both are equal.

Practice task:

$$g(x) = \begin{cases} x + 1, & x < 3 \\ 7, & x \geq 3 \end{cases}$$

Find the left limit, right limit, function value, and continuity at $x = 3$.

Student: Left limit is 4. Right limit is also 4, because x is still approaching 3. Function value is 7. I think maybe it is continuous because $g(3)$ exists.

Mentor: This is the rule-choice mistake. From the right side, the formula is not $x + 1$; it is the constant 7. Also, having $g(3)$ exist is not enough for continuity. Try again.

Student: Left limit: $3 + 1 = 4$. Right limit: 7. Function value: $g(3) = 7$. Since $4 \neq 7$, the two-sided limit does not exist, so the function is not continuous.

Mentor: Correct. Transfer check: if the second piece were 4 instead of 7, would it be continuous?

Student: Yes. Left limit would be 4, right limit would be 4, and $g(3) = 4$.

Mentor: Passed.

Lesson 3: Derivative as Rate of Change

Mentor: Video: Khan Academy, "Derivatives introduction": <https://www.youtube.com/watch?v=N2PpRnFqnqY>

Short summary: a derivative is instantaneous rate of change. It is the limit of average rates of change:

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Practice task: compute $f'(3)$ for $f(x) = x^2$ using the definition.

Student: $f(3+h) = (3+h)^2 = 9 + 6h + h^2$, and $f(3) = 9$. So

$$\frac{f(3+h) - f(3)}{h} = \frac{6h + h^2}{h} = 6 + h$$

As $h \rightarrow 0$, $f'(3) = 6$.

Mentor: Good. What does **6** mean if $f(x)$ is position in meters and x is time in seconds?

Student: At 3 seconds, the position is changing at **6** meters per second.

Mentor: Transfer check: what would $f'(a)$ be for $f(x) = x^2$?

Student: $(a + h)^2 - a^2 = 2ah + h^2$. Divide by h : $2a + h$. Limit is $2a$.

Mentor: Passed. You can connect the formula to rate of change.

Lesson 4: Power Rule

Mentor: Video: Khan Academy, "Taking derivatives": <https://www.youtube.com/watch?v=rAof9Ld5sOg>

Short summary: the power rule says

$$\frac{d}{dx}x^n = nx^{n-1}$$

Constant multiples stay in front, and constants have derivative 0.

Practice task: differentiate $y = 5x^4 - 3x^2 + 8$, then find the slope at $x = -1$.

Student: $y' = 20x^3 - 6x + 8$. At $x = -1$, that gives $-20 + 6 + 8 = -6$.

Mentor: You made the common constant mistake. The derivative of 8 is 0, not 8. Correct the derivative and the slope.

Student: Right. The derivative is

$$y' = 20x^3 - 6x$$

At $x = -1$:

$$20(-1)^3 - 6(-1) = -20 + 6 = -14$$

Mentor: Correct. Transfer check: differentiate $7\sqrt{x}$.

Student: Rewrite as $7x^{1/2}$. Then

$$\frac{7}{2}x^{-1/2} = \frac{7}{2\sqrt{x}}$$

Mentor: Passed.

Lesson 5: Product and Quotient Rules

Mentor: Video: Khan Academy, "Product rule proof": <https://www.youtube.com/watch?v=L5ErIC0COxl>

Short summary: use product rule for multiplication and quotient rule for division:

$$(uv)' = u'v + uv'$$

$$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$$

Practice task:

$$y = \frac{2x + 5}{x^2 + 1}$$

Set up y' using the quotient rule.

Student: I practiced this after the video. It was fine. I just used the quotient rule.

Mentor: That is not evidence. What exactly were u , v , u' , and v' ?

Student: Okay, I did not actually write it out. Let me do it now. $u = 2x + 5$, so $u' = 2$. $v = x^2 + 1$, so $v' = 2x$. Therefore,

$$y' = \frac{2(x^2 + 1) - (2x + 5)(2x)}{(x^2 + 1)^2}$$

Mentor: Good. That is the bluff-catching moment: I did not accept "I practiced" without details. Transfer check: why is the sign in the numerator minus, not plus?

Student: Because quotient rule is $u'v - uv'$. Plus would be product rule behavior.

Mentor: Passed.

Lesson 6: Chain Rule

Mentor: Video: Khan Academy, "Chain rule": <https://www.youtube.com/watch?v=0T0QrH056qg>

Short summary: for a composite function, differentiate the outside and multiply by the derivative of the inside.

Practice task:

$$y = \sqrt{4x^3 + 2}$$

Name the inner function and differentiate.

Student: Inner function is $4x^3 + 2$. Since square root is power $1/2$,

$$y' = \frac{1}{2}(4x^3 + 2)^{-1/2}$$

Mentor: You found the outer derivative, but you forgot the inner derivative. What is $\frac{d}{dx}(4x^3 + 2)$?

Student: It is $12x^2$. Corrected:

$$y' = \frac{1}{2}(4x^3 + 2)^{-1/2} \cdot 12x^2 = \frac{6x^2}{\sqrt{4x^3 + 2}}$$

Mentor: Correct. Transfer check: differentiate $(3x^2 - 1)^5$.

Student: Outer derivative gives $5(3x^2 - 1)^4$. Inner derivative is $6x$, so

$$30x(3x^2 - 1)^4$$

Mentor: Passed.

Lesson 7: Implicit Differentiation

Mentor: Video: Khan Academy, "Implicit differentiation": <https://www.youtube.com/watch?v=5pPdtHF1Bhw>

Short summary: when y is not isolated, treat y as a function of x . Differentiating y^3 gives $3y^2 \frac{dy}{dx}$.

Practice task:

$$x^2y + y^3 = 10$$

Differentiate and solve for $\frac{dy}{dx}$.

Student: Derivative of x^2y is $2xy + x^2 \frac{dy}{dx}$. Derivative of y^3 is $3y^2$. So

$$2xy + x^2 \frac{dy}{dx} + 3y^2 = 0$$

Mentor: Product rule on x^2y was good, but y^3 needs chain rule because y depends on x . Try that term again.

Student: It should be $3y^2 \frac{dy}{dx}$. Then

$$2xy + x^2 \frac{dy}{dx} + 3y^2 \frac{dy}{dx} = 0$$

Group terms:

$$\frac{dy}{dx}(x^2 + 3y^2) = -2xy$$

So

$$\frac{dy}{dx} = \frac{-2xy}{x^2 + 3y^2}$$

Mentor: Correct. Transfer check: what earlier rule did you use in x^2y ?

Student: Product rule, because x^2 and y are multiplied.

Mentor: Passed.

Lesson 8: Tangent Lines and Linear Approximation

Mentor: Video: Khan Academy, "Derivative as slope of a tangent line":
<https://www.youtube.com/watch?v=ANyVpMS3HL4>

Short summary: a tangent line at $x = a$ uses the point and derivative:

$$y - f(a) = f'(a)(x - a)$$

It can approximate nearby function values.

Practice task: find the tangent line to $f(x) = x^3 - x$ at $x = 2$, then approximate $f(2.1)$.

Student: $f(2) = 8 - 2 = 6$. $f'(x) = 3x^2 - 1$, so $f'(2) = 11$. Tangent line:

$$y - 6 = 11(x - 2)$$

So $y = 11x - 16$. At $x = 2.1$, the approximation is 7.1.

Mentor: Good. Now compare with the exact value.

Student: $f(2.1) = 2.1^3 - 2.1 = 9.261 - 2.1 = 7.161$. The tangent estimate is close and a little low.

Mentor: Why might it be low?

Student: Since $f''(x) = 6x$, near $x = 2$ the graph is concave up, so the tangent line lies below the curve.

Mentor: Passed. You used both derivative and interpretation.

Lesson 9: Optimization

Mentor: Video playlist: Khan Academy derivative applications / optimization:
<https://www.youtube.com/playlist?list=PLSQL0a2vh4HDHy9nIoEb-iQ9eQVYoXiHy>

Short summary: optimization means turning a word problem into a one-variable function, differentiating, setting the derivative equal to zero, and checking whether the result is a maximum or minimum.

Practice task: a rectangle has perimeter 40. What dimensions maximize its area?

Student: $2l + 2w = 40$, so $l + w = 40$, and $w = 40 - l$. Then

$$A = l(40 - l) = 40l - l^2$$

Set $A' = 40 - 2l = 0$, so $l = 20$, $w = 20$.

Mentor: Check the perimeter equation. If $2l + 2w = 40$, what is $l + w$?

Student: I divided wrong. $2l + 2w = 40$ means $l + w = 20$, so $w = 20 - l$.

$$A = l(20 - l) = 20l - l^2$$

Then $A' = 20 - 2l$. Set $A' = 0$, so $l = 10$, $w = 10$. Maximum area is 100.

Mentor: How do you know it is a maximum?

Student: $A'' = -2$, so the parabola opens downward and the critical point is a maximum.

Mentor: Passed.

Lesson 10: Related Rates

Mentor: Video: Khan Academy, "Related rates intro": <https://www.youtube.com/watch?v=kQF9pOqmS0U>

Short summary: related rates connect quantities that change over time. Write an equation, differentiate both sides with respect to time, substitute known values, then interpret sign and units.

Practice task: a 10 ft ladder leans against a wall. The bottom slides away at 1 ft/s. How fast is the top sliding down when the bottom is 6 ft from the wall?

Student: Let x be the bottom distance and y be the height. Then

$$x^2 + y^2 = 100$$

Differentiate:

$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$$

At $x = 6$, $y = 8$, and $\frac{dx}{dt} = 1$:

$$12 + 16 \frac{dy}{dt} = 0$$

So $\frac{dy}{dt} = \frac{3}{4}$ ft/s.

Mentor: The magnitude is right, but the sign is wrong. If the bottom moves away, the top moves down, so $\frac{dy}{dt}$ must be negative. Solve the last step again.

Student: $16 \frac{dy}{dt} = -12$, so

$$\frac{dy}{dt} = -\frac{12}{16} = -\frac{3}{4}$$

The top slides down at 0.75 ft/s.

Mentor: Final transfer check: differentiate

$$y = \frac{(x^2 + 1)^3}{x - 1}$$

This combines quotient rule and chain rule.

Student: $u = (x^2 + 1)^3$, so $u' = 6x(x^2 + 1)^2$. $v = x - 1$, so $v' = 1$. Therefore,

$$y' = \frac{6x(x^2 + 1)^2(x - 1) - (x^2 + 1)^3}{(x - 1)^2}$$

Mentor: Course complete. You corrected errors, handled a bluff honestly, and transferred old rules to a new problem.

5. Rationale and Self-Evaluation

At first I tried to make the course without videos: the mentor would simply explain each lesson and then quiz the student. That version had a problem. Because both characters were language models, the student sometimes used knowledge the mentor had not taught yet. For example, it could suddenly mention L'Hopital's rule, integration, or other calculus ideas even though the course had not reached them. That made it harder to tell whether the student learned from the mentor or already "knew" the idea because of the model's pretraining.

I also had problems with the first raw multi-agent runs. The mentor sometimes accepted vague answers too easily, repeated the same limits question instead of moving on, or skipped the teaching step and jumped straight to examples. In some runs the student referred to "earlier" work even when it was supposed to be the first message of a fresh dialogue. The small local models also made arithmetic mistakes and occasionally failed to catch each other's errors.

I changed the design to a video-based course to make the teaching source clearer. Each lesson now begins with a YouTube or Khan Academy link, followed by a short summary, a practice task, a mistake or bluff from the student, a mentor correction, and a transfer check. This gives the mentor a more concrete basis for asking "did you apply the lesson?" rather than just asking whether the student understood.

I also revised the prompts after reading the persona-prompting paper. The paper argues that persona prompts are task- and model-dependent: they can help alignment and formatting, but can damage factual or reasoning accuracy. Because calculus needs accuracy more than roleplay, I removed names and heavy persona framing from the prompts. The final prompts are shorter and task-focused: teach, ask for evidence, catch mistakes, and verify transfer.

The strongest verification moment is Lesson 5. The student says she practiced the quotient rule, but the mentor asks for u , v , u' , and v' . A student who really practiced can provide those details; a bluffing student cannot. The mentor catches the bluff and refuses to move on until the student sets up the derivative.

The transcript in the final PDF is a cleaned version of the experiment rather than a verbatim raw model log. I did edit it: I removed drift, made the lesson lengths even, and made the dialogue match the final video-lesson format. The raw model logs are still kept separately,

but the submitted transcript is cleaned so the homework reads coherently and honestly documents the problems I ran into.

The remaining weakness is that conversation alone still cannot prove that the student watched the videos or worked independently. The mentor can reduce this risk by asking for steps, mistakes, transfer tasks, and explanations, but it cannot eliminate it completely.